Introduction to Data Science

데이터사이언스개론

Week 2

최예림

서울여자대학교 데이터사이언스학과

yerim.choi@swu.ac.kr

강의 일정

week	date	topic	assignment	practice
1	03/02/2023	강의 및 학과 소개		
2	03/09/2023	데이터 활용하기 1: 데이터사이언스의 개념	과제 1 공지	
3	03/16/2023	데이터 활용하기 2: 야생멧돼지와 데이터 Part 1		
4	03/23/2023	데이터 활용하기 3: 야생멧돼지와 데이터 Part 2		
5	03/30/2023	데이터 활용하기 4: 패션과 데이터		
6	04/06/2023	데이터 활용하기 5: 삼성과 데이터 Part 1	과제 1 제출 기한	
7	04/13/2023	데이터 활용하기 6: 삼성과 데이터 Part 2		
8	04/20/2023	중간고사		
9	04/27/2023	데이터 이해하기 1: 데이터사이언스의 개념 Revisited	과제 2 공지	
10	05/04/2023	데이터 이해하기 2: 데이터 찾아보기		
11	05/11/2023	데이터 이해하기 3: 데이터 탐색하기		실습 1 데이터 수집
12	05/18/2023	데이터 이해하기 4: 데이터 분석하기		실습 2 데이터 탐색
13	05/25/2023	데이터 이해하기 5: 데이터 보여주기		실습 3 데이터 분석
14	06/01/2023	과제 2 발표 Part 1	과제 2 제출 기한	실습 4 발표 1
15	06/08/2023	과제 2 발표 Part 2		실습 5 발표 2

토론 조

(토론) 자기소개

이름, 장래희망, 특장점, DS에 지원한 동기(연계전공을 하는 이유), …

과제 1 공지

- 목표 : 이력서(지금까지의 이력 + 졸업까지 달성하고 싶은 이력* 포함)를 작성하여 지금의 나를 이해하고, 앞으로의 목표를 수립한다.
  *지금까지 이력과 졸업까지 달성하고 싶은 이력이 구분되도록 작성
- 양식 : 자유 양식(다양한 이력서를 많이 찾아보세요!)
  *MS Word 또는 한글로 작성 후 PDF로 변환하여 제출
- 파일 제목 : (DS) 과제 1 이름
- Due : 6주차 강의시작 전까지 hard copy(강의실 앞) 및 soft copy(이클래스) 제출
- 비중 : 5%

- 참고자료
  - https://ybson.blog/

강의목표

- 데이터사이언스의 개념을 이해한다.
- 데이터사이언스 과정을 이해한다.

- Keywords
  - Data science
  - Data and big data
  - Data preprocessing
  - Data visualization
  - Machine learning & deep learning

Data Science

Data Science

- Data science is an inter-disciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data and apply knowledge and actionable insights from data across a broad range of application domains

- Keywords
  - Data
  - Knowledge & insight
  - Application, inter-disciplinary
  - Scientific methods …

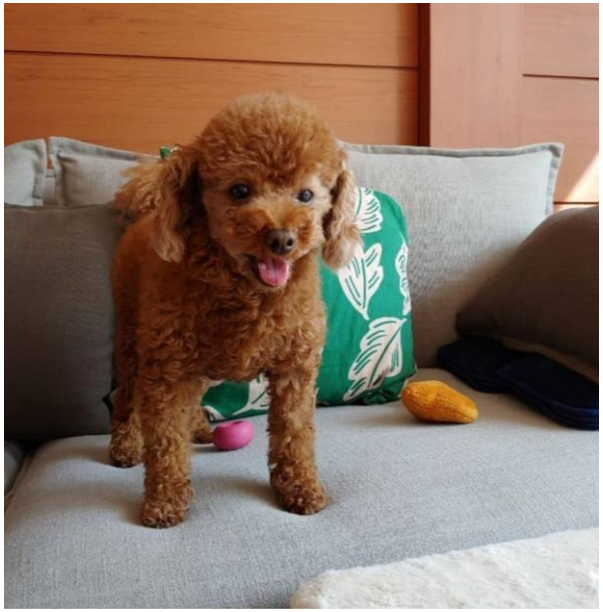
Data

- Symbols or signs, <u>representing</u> stimuli or signals, that <u>represent</u> real-worlds

- Known as …
  - Economic goods, 21세기의 원유

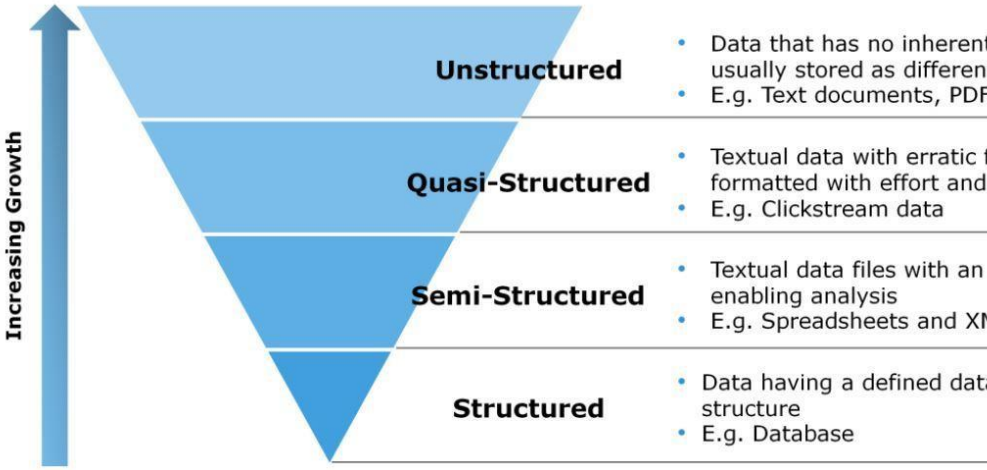
Data Example

- Student data
  - Name, student number, major, year, grade, …
- Review logs
  - User, content, datetime, rating, comment, image
- Dog images
- E-mail texts



Data Types

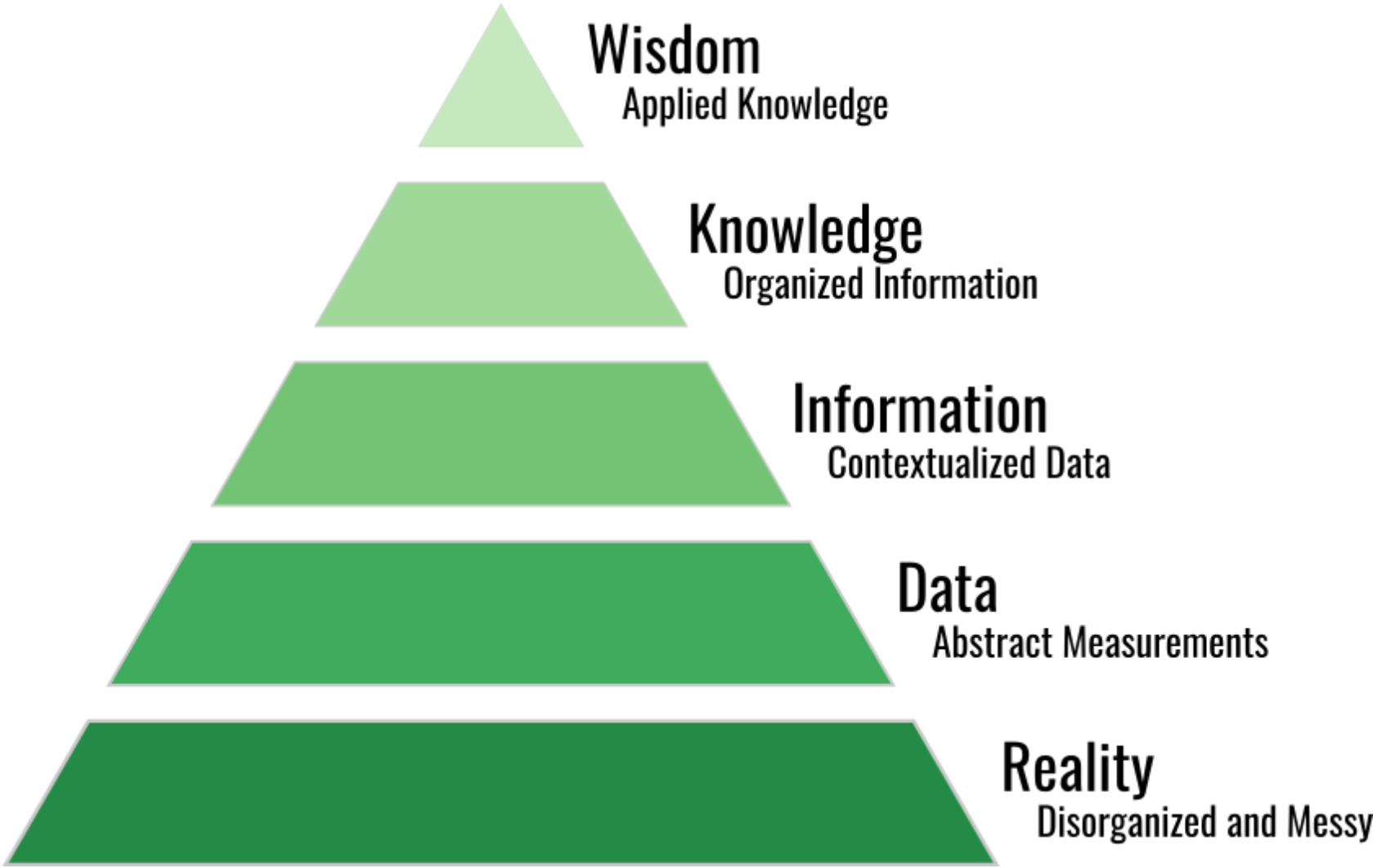
- Structured data
  - 구조가 정의되어 있는 데이터
  - e.g., student data, review logs



- Unstructured data
  - 구조가 정의되어 있지 않은 데이터
  - e.g., images, text

- Semi-structured data
  - 구조가 정의되어 있지만, 유연하게 변경이 가능한 데이터
  - XML, json

Data : DIKW Model



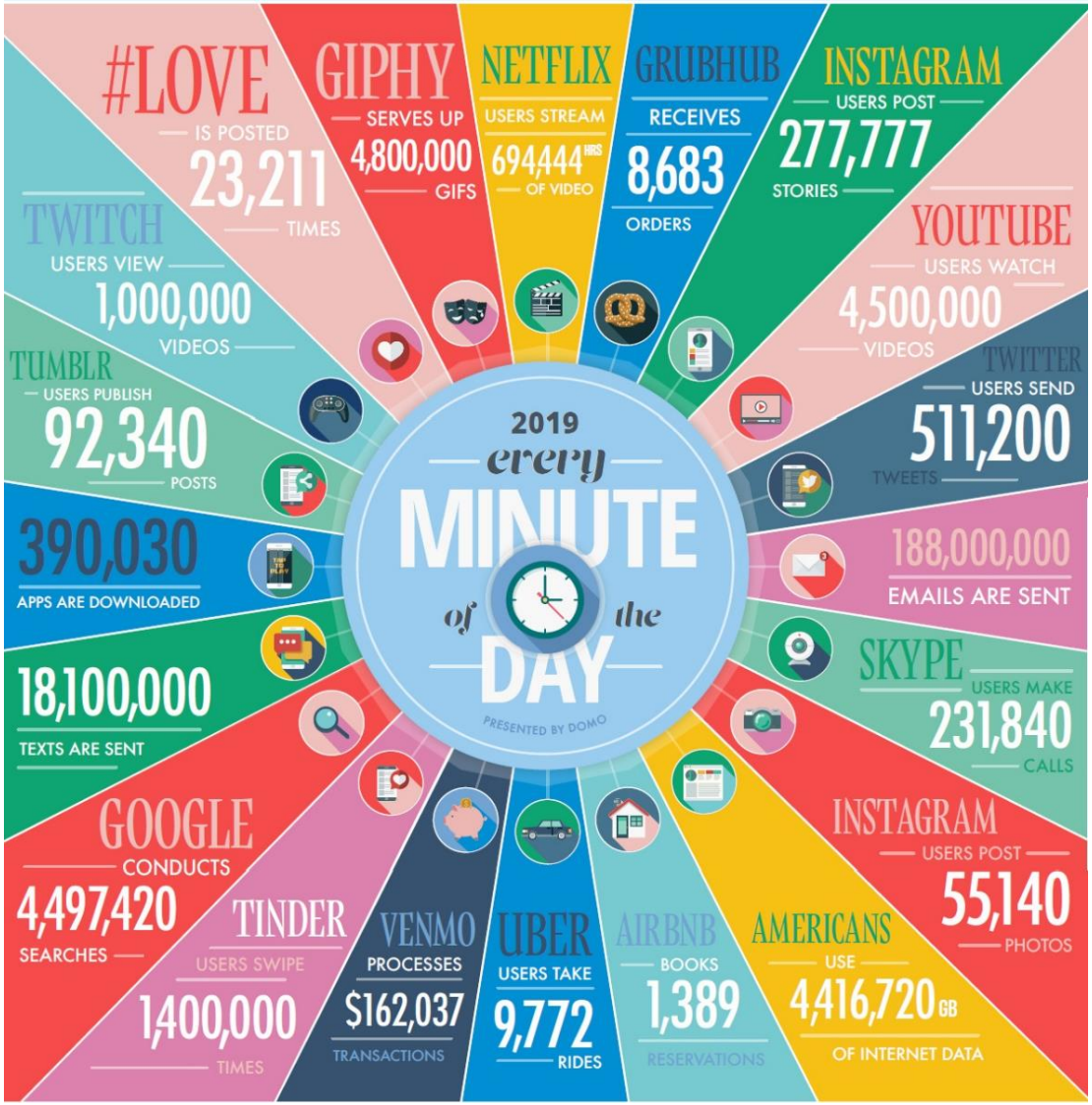
Big Data

- 3Vs
  - Vast mount of data
  - Variety data
  - High velocity

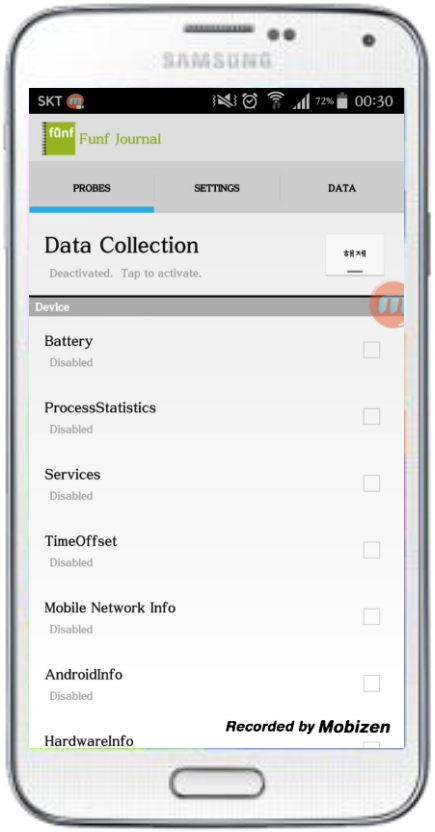
- Thanks to
  - Sensor technology
  - IoT
  - Networks
  - Data storage

https://www.visualcapitalist.com/big-data-keeps-getting-bigger/

Big Data



Big Data(you!)



Funf Journal

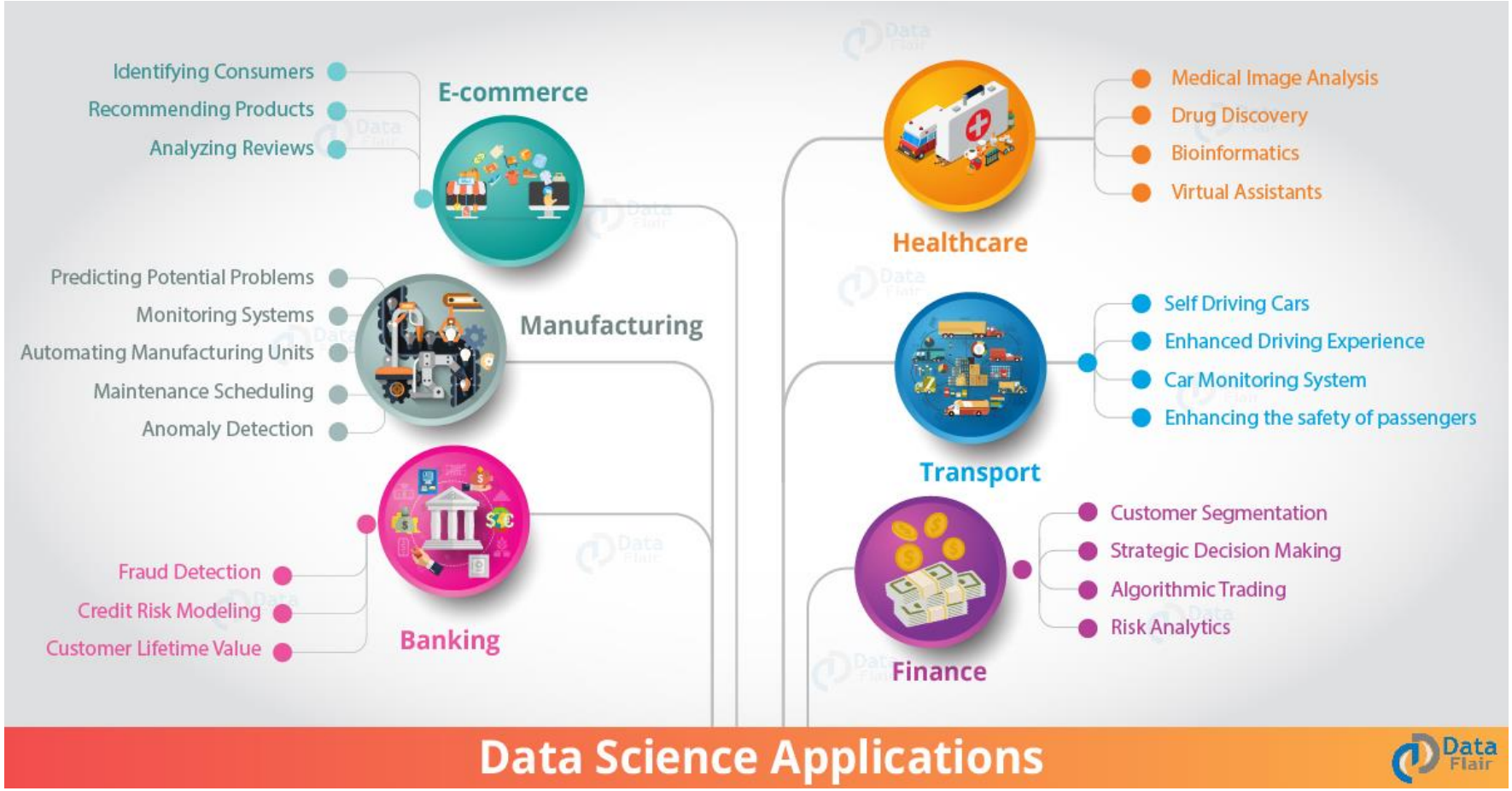
Device	Battery Hardware Info Telephony
Environment	Light Sensor Proximity Sensor Magnetic Field Sensor Pressure Sensor Temperature Sensor
Positioning	Location Bluetooth Wifi Cell
Motion	Accelerometer Sensor Gravity Sensor Gyroscope Sensor Orientation Sensor Activity
Device Interaction	Applications Screen Browser Bookmarks Browser Searches Videos Audio Files Images
Social	Contact Call Log SMS

(토론) 내가 생산하고 있는 데이터는?

- 각 개인이 만들어내고 있는 데이터에는 어떤 것이 있을까?

https://data-flair.training/blogs/data-science-applications/

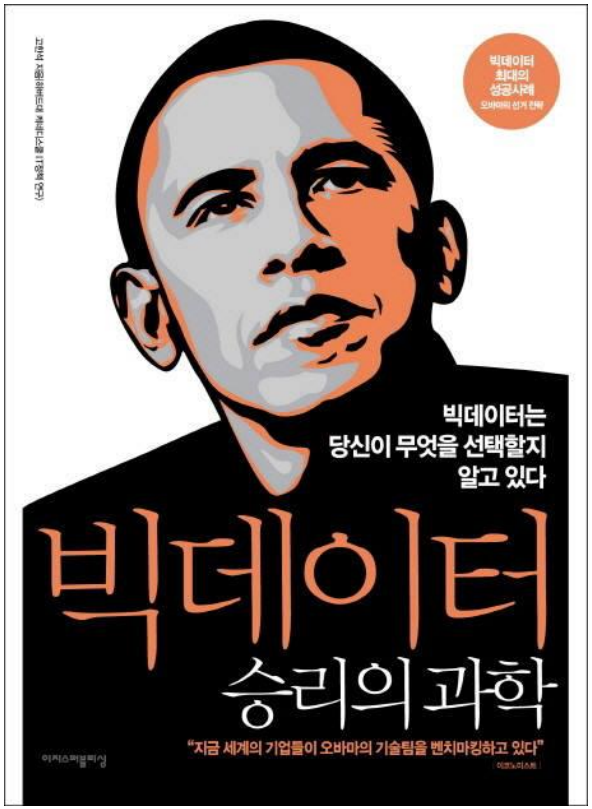
Applications with Knowledge and Insight



https://www.linkedin.com/pulse/becoming-netflix-intelligent-something-every-company-can-schmarzo/?utm_content=74790724&utm_medium=social&utm_source=twitter

Applications with Knowledge and Insight

Applications with Knowledge and Insight



선거 2년 전부터 빅데이터를 수집·분석하여 로드맵 도출

짐 메시나, 시카고 선거캠프 본부장

"우리는 선거운동의 일거수일투족을 수치화 했다"

빅데이터 분석을 통한 맞춤형 지지 호소 및 어필

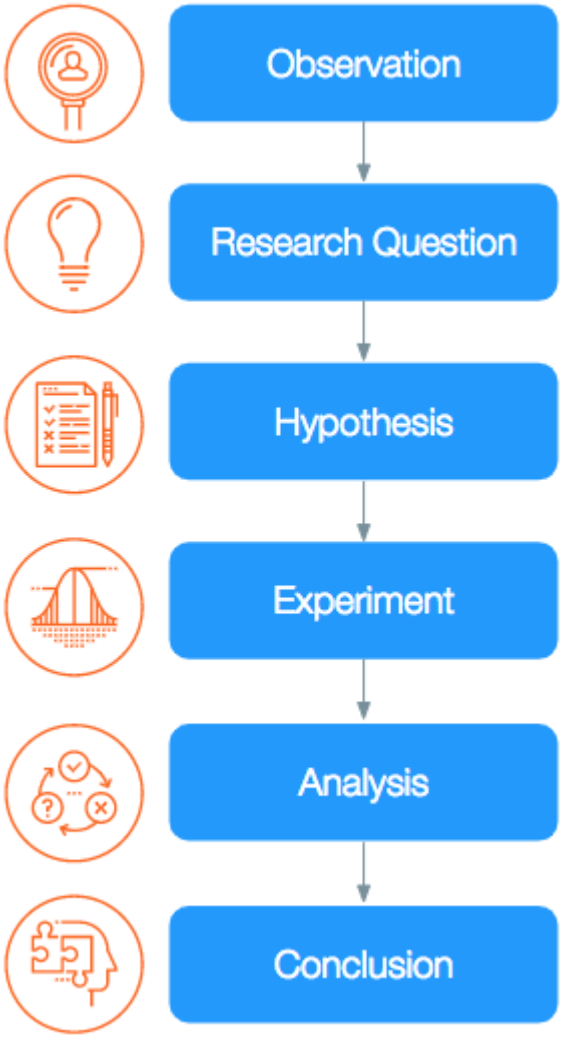
레이드 가니, 빅데이터 분석 전문가

"유권자를 철저히 분석해 특정 그룹에 어떤 방식으로 접근해야 하는지 세부전략 만들어 보고"

Data Science Methods

- Scientific methods

- Key characteristics
  - Purposiveness
  - Objectivity
  - Replicability
  - Reliability
  - Validity
  - Rigor
  - Testability and **generality**



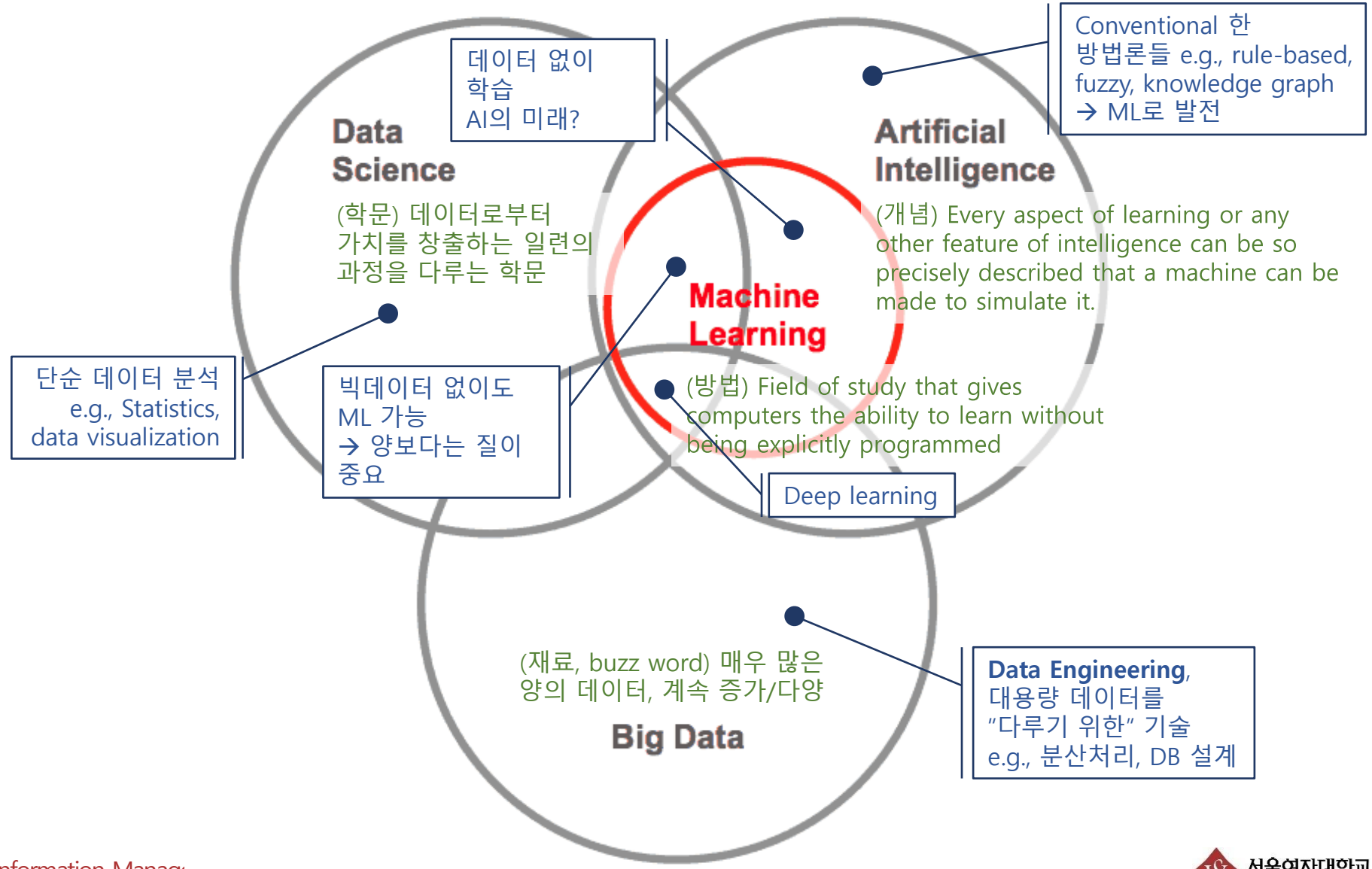
https://theintactone.com/2019/03/03/brm-u1-topic-5-characteristics-of-scientific-method/
https://engineering.indeedblog.com/blog/2018/12/qualitative-methods-for-better-data-science/

Data Science Methods

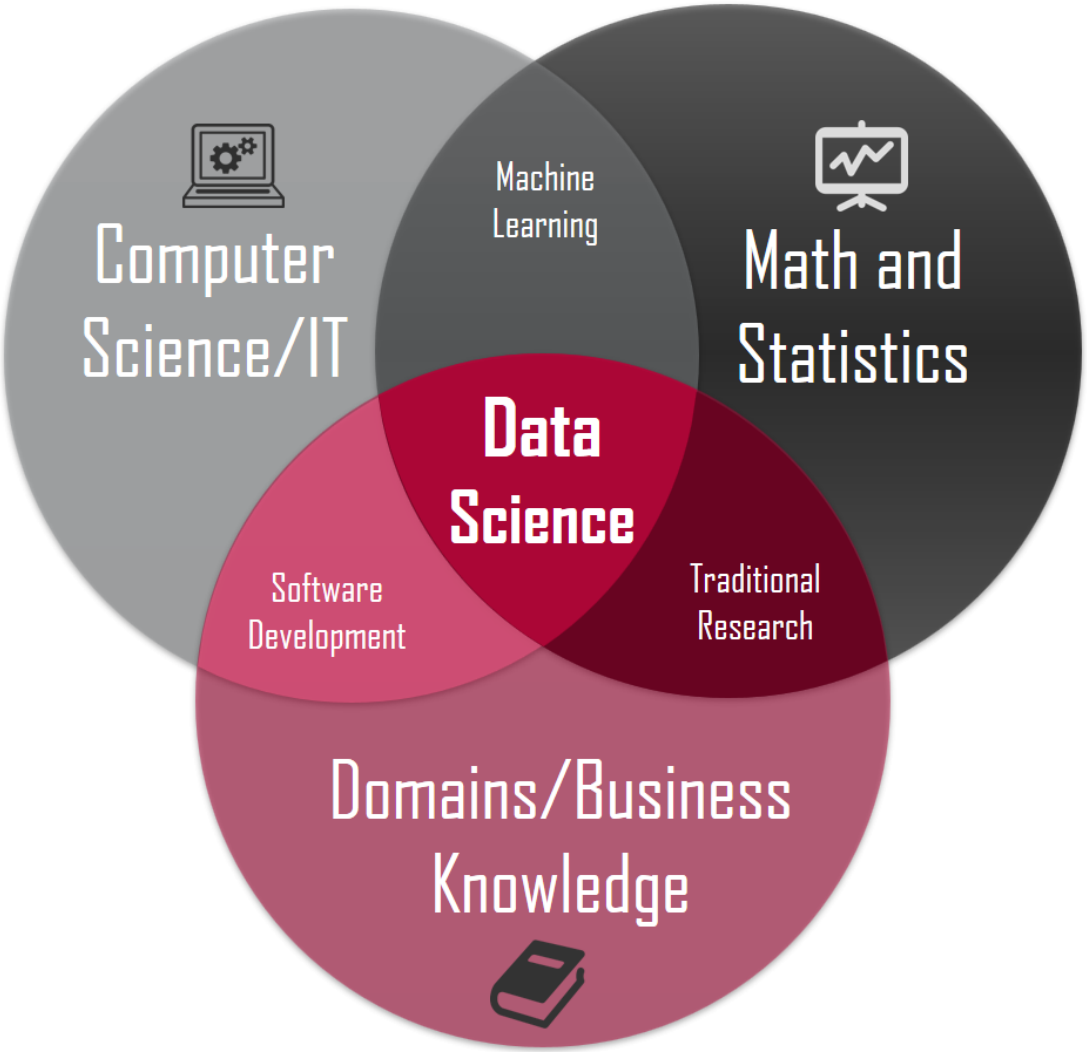
Domain	Category	Examples
Business Intelligence + Business Analytics	**Descriptive**	• Scorecards • Descriptive Statistics, e.g. Mean, Variance, Range, Confidence Intervals • Histograms, Scatterplots, Pareto Charts, Time Series, Boxplots, Pie/Waffle Charts • Correlation and Association Analysis • Signal to Noise Analysis
	Diagnostic or Explanatory	• Design of Experiments • Root Cause Analysis • Clustering and Community Detection • Text Mining • Network Analysis • Data Reduction (e.g. PCA, SVD, Factor/Discriminant Analysis)
Data Science + Operations Research	**Predictive**	• Regression • Classification • Neural Networks • Time Series Forecasting • Statistical Process Control (SPC) • Scoring Systems • Risk Assessment • Ensemble Methods, e.g. Random Forest
	Prescriptive	• Simulations • Optimization Techniques • Decision Trees • Discrete Event Simulation (DES) • Simulation-based Scenario Analysis • Recommender Systems • A/B Testing (and multi-armed bandit) • Multi-criteria Decision Making • Reinforcement Learning • System Dynamics

Table 1. Examples of methods to generate descriptive, diagnostic/explanatory, predictive, and prescriptive analytics.

Data Science vs. …



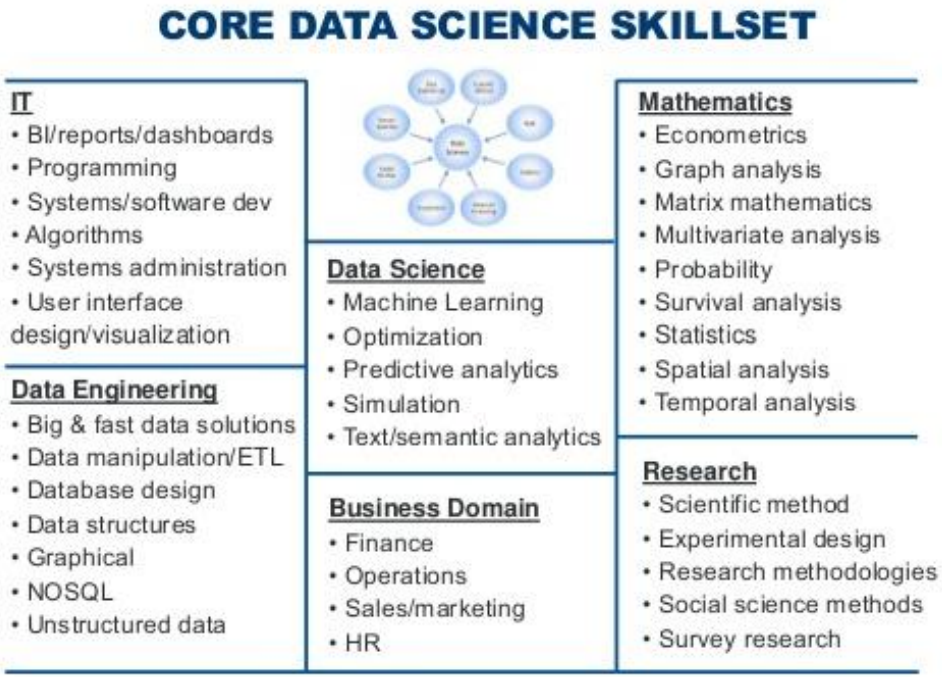
필요한 역량



Source : Michael Barber, Data science concepts you need to know! , 2018.

필요한 역량

- **Mathematics**
  - Statistics and probability
  - Linear algebra
  - Optimization
  - Information theory
- **Computer science**
  - Computer structure
  - Data structure, algorithm
  - GUI & interface
  - Programming languages : Python, R
  - Tools : Git, Docker, …
- **Data engineering & analysis**
  - Database, big data platform
  - Machine learning, deep learning
  - Data visualization
- **Applications & ethics**
  - Domain knowledge & project management
  - Rules, ethics, security



필요한 역량 : Tools

Data Store
Transform
Model
Visualize
Other Tools
mongoDB
MySQL
Spark
SQL
scikit learn
pandas
Spark
ggplot2
D3
kafka
Luigi
Joins are what RDBMS's do for a living
If you change the way you look at things, the things you look at change.
AI usually beats natural stupidity
Dont trust everything you see, even salt looks like sugar
I have enough tools, said no data scientist ever

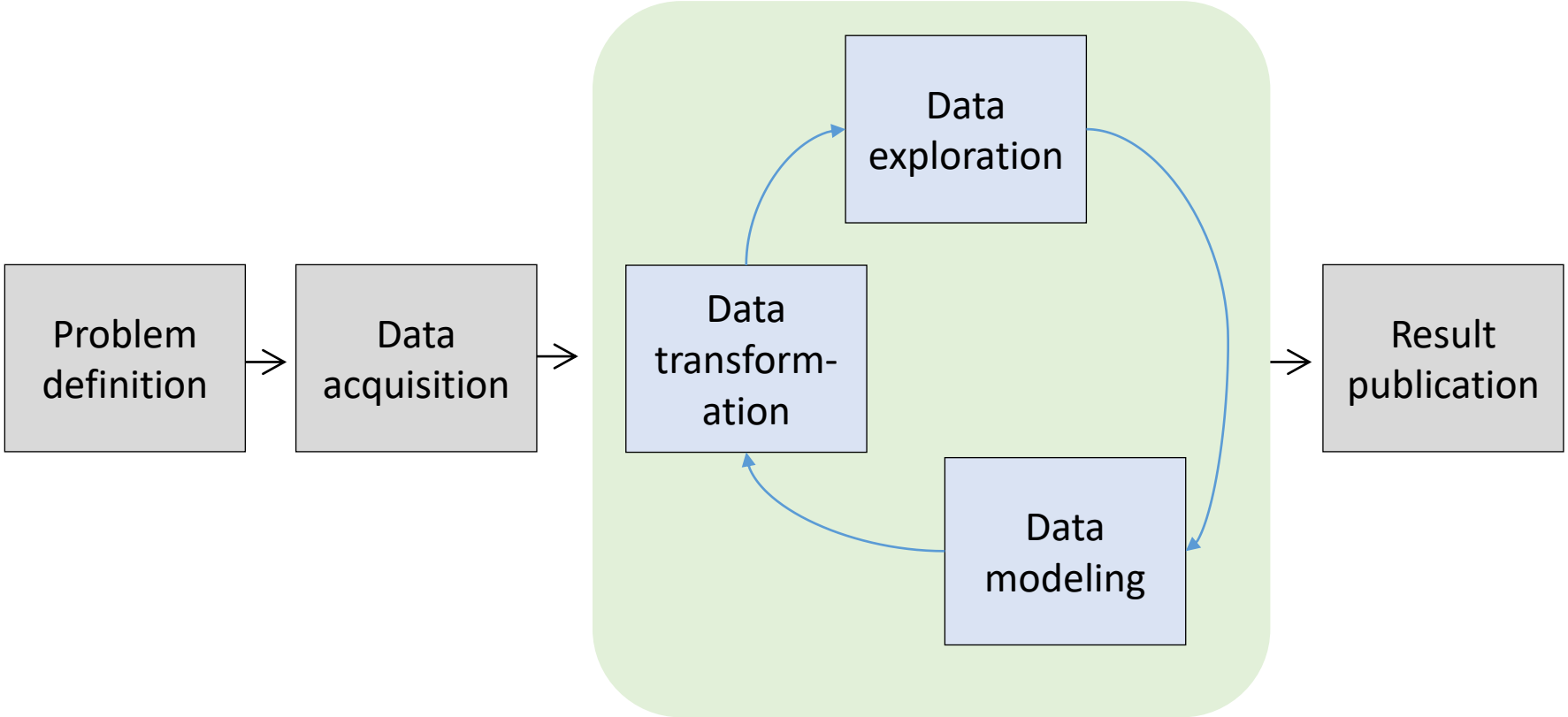
(토론) 내게 가장 필요한 DS 역량은?

- 본인이 가장 잘하거나 부족할 것이라고 생각하는 DS 역량과 그 이유를 토론하시오.

- 발표
  - 가장 특이한 케이스 한 명의 내용을 함께 공유

Data Science Process

Data Science Pipeline



Data Science Pipeline

- **Problem definition**
  - Understanding domain
  - Numerically define problem
- **Data acquisition**
  - Given data, open datasets
  - Data production : data entry, data annotation
  - Web crawling
- **Data transformation**
  - Noise removal / outlier detection
  - Join / aggregation
  - Feature engineering / embedding / vectorization
  - Dimension reduction
- **Data exploration**
  - Data summary / correlation analysis
  - Visualization
- **Data modeling**
  - Descriptive : rule mining, clustering
  - Predictive : regression, classification
- **Result publication**
  - Reporting : infographic
  - Software : API

Summary

- Data science는 data로부터 insight를 도출하고, 그 결과를 적용하는 과정 및 방법론을 다루는 다학제적 학문
  - DIKW pyramid
  - Big data : 3Vs
  - Unstructured, structured data

- Data science는 문제 정의 > 데이터 확보 > 데이터 변환 > 데이터 관찰 > 데이터 모델링 > 결과 활용의 단계를 거침

Data Acquisition

Data acquisition

- Collecting data to address the given problem

- Meaningfulness of data
  - Able to describe the problem situation e.g. statistics

- Size of data
  - Small number of data → bias(overfitting), training x
  - Too large → impossible to store → preprocessing & store

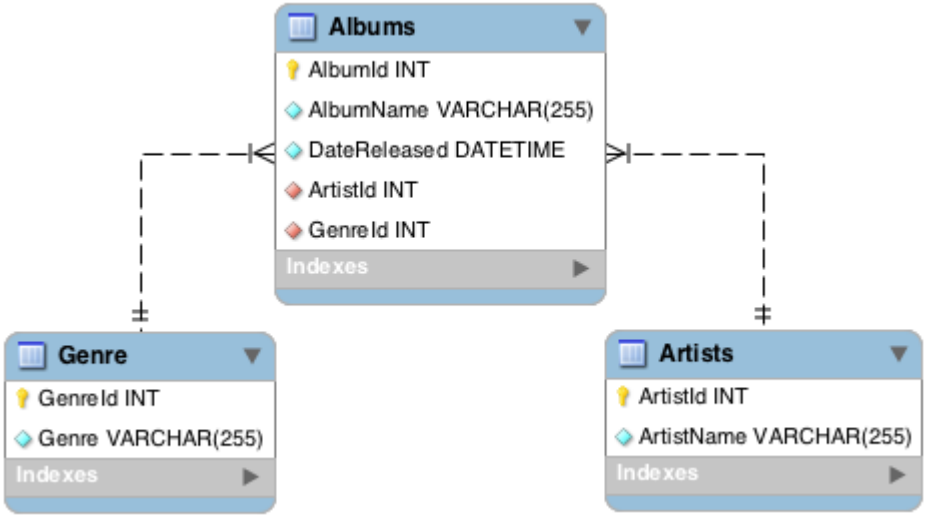
- Real vs. experimental
  - Meaningful : real > experimental
  - Label : real < experimental
  - Noise : real < experimental

Open datasets

- Lots of open datasets
  - 모델 pretrain을 위한 데이터로 사용 가능
- Traditional
  - ML Repository : http://archive.ics.uci.edu/ml/index.php
  - MNIST : http://yann.lecun.com/exdb/mnist/
- Domestic
  - AI hub : http://www.aihub.or.kr/
  - 공공데이터포털 : https://www.data.go.kr/
- Others
  - Open Images Dataset : https://opensource.google.com/projects/open-images-dataset
  - Fashion-MNIST : https://github.com/zalandoresearch/fashion-mnist
  - Kaggle : https://www.kaggle.com/

Data basics

- Data size
  - 건, 용량(GB)
  - # of instances
  - # of attributes



- Data schema (meta info.)
  - The skeleton structure that represents the logical view of the entire database(*source: https://www.tutorialspoint.com/dbms/dbms_data_schemas.htm)
  - Attribute list and types
  - Relation between tables
  - Database → set of tables

Data types

Data Type			Possible values	Example usage
Categorical	binary		0, 1 (arbitrary labels)	binary outcome ("yes/no", "true/false", "success/failure", etc.)
	categorical		1, 2, ..., K (arbitrary labels)	categorical outcome (specific blood type, political party, word, etc.)
	ordinal		integer or real number (arbitrary scale)	relative score, significant only for creating a ranking
Numerical	discrete		nonnegative integers (0, 1, ...)	number of items (telephone calls, people, molecules, births, deaths, etc.)
	continuous	real-valued additive	real number	temperature, relative distance, location parameter, etc. (or approximately, anything not varying over a large scale)
		real-valued multiplicative	positive real number	price, income, size, scale parameter, etc. (especially when varying over a large scale)

➔ Categorical & Numeric

Data summary

- Numerical
  - Most statistics
  - Mean, median, std., max/min, quartile
  - Correlation

- Categorical
  - Frequencies
  - Mode
    - The mode of a set of data values is the value that appears most often
  - Cooccurrence

Data types (format)

- Structured data
  - Has a predefined data format e.g. matrix (instance x feature)
  - Relational database

- Unstructured data
  - No predefined data format
  - Examples
    - Text data
    - Image data
    - Sequential data
    - Network data

- Semi-structured data
  - Has changeable data format e.g. Json, XML

Unstructured data

- Text data
  - Composed of strings
  - Sometimes long document
  - Word vector, w2v

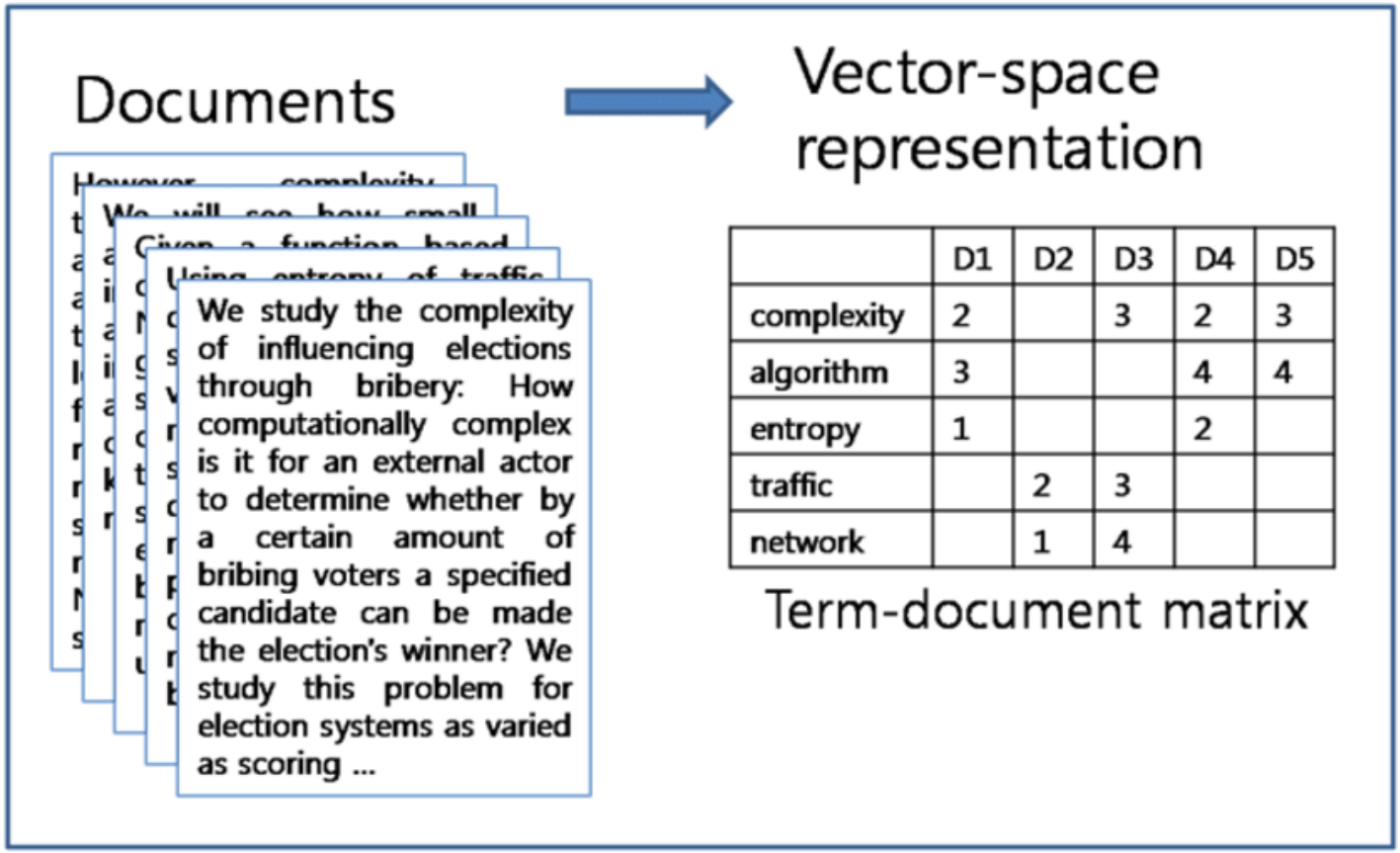
- Image / video data
  - List of vectors of (R, G, B) indicating each pixel

- Sequential data
  - Data collected according to time t
  - Signal data, audio signal, brain signal
  - Aggregated features e.g. mean/std. over 5min.
  - Few models deal with sequences e.g. HMM, RNN

- Network data
  - Node-edge
  - Adjacent, hop

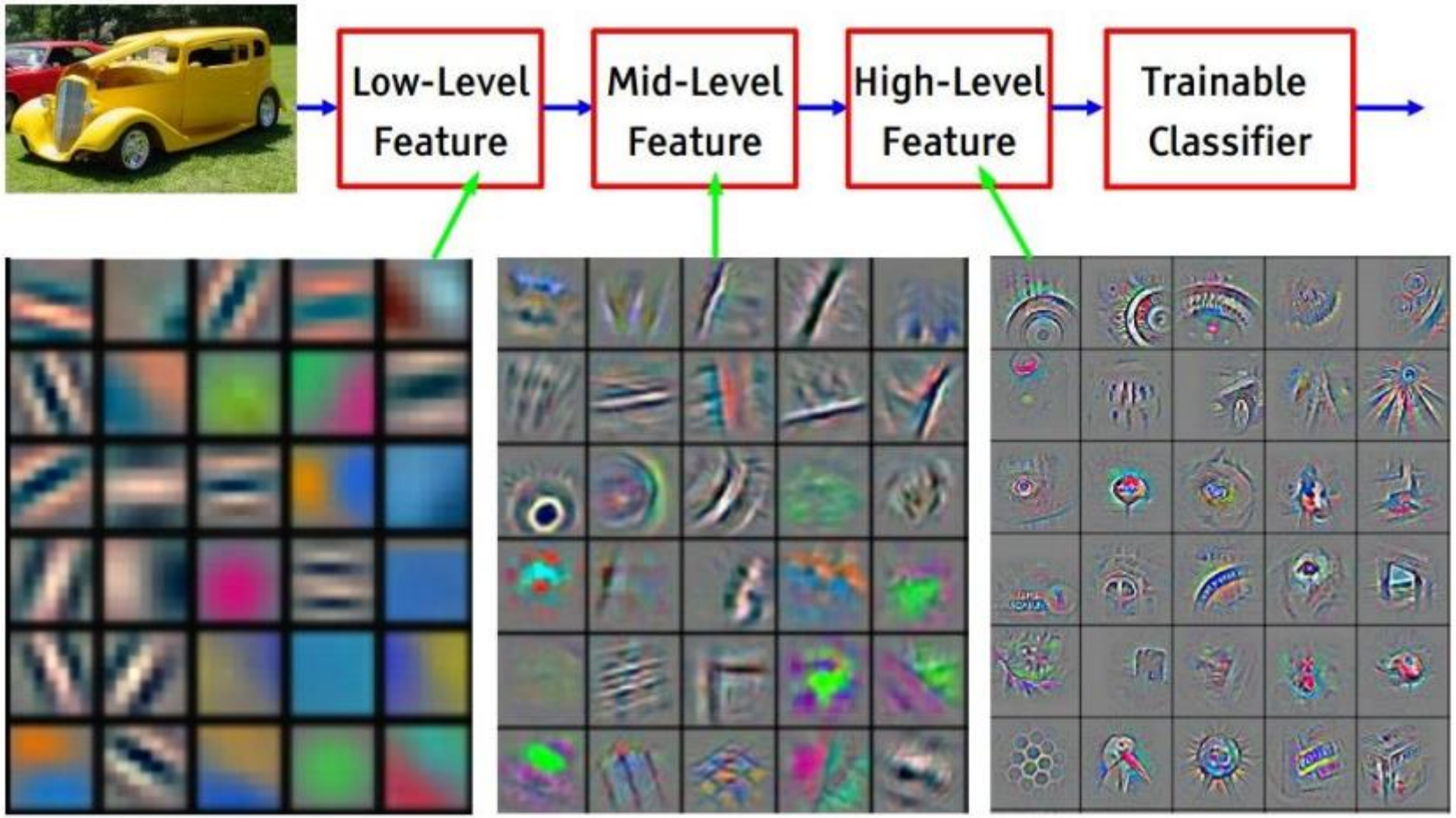
Text data

- Document classification



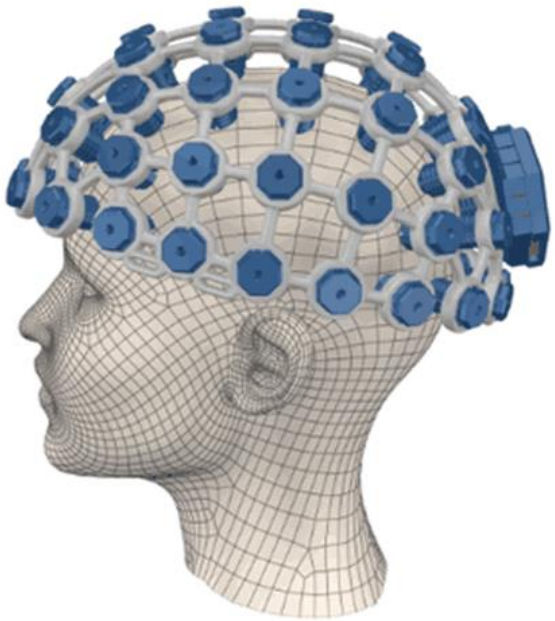
	D1	D2	D3	D4	D5
complexity	2		3	2	3
algorithm	3			4	4
entropy	1			2	
traffic		2	3		
network		1	4		

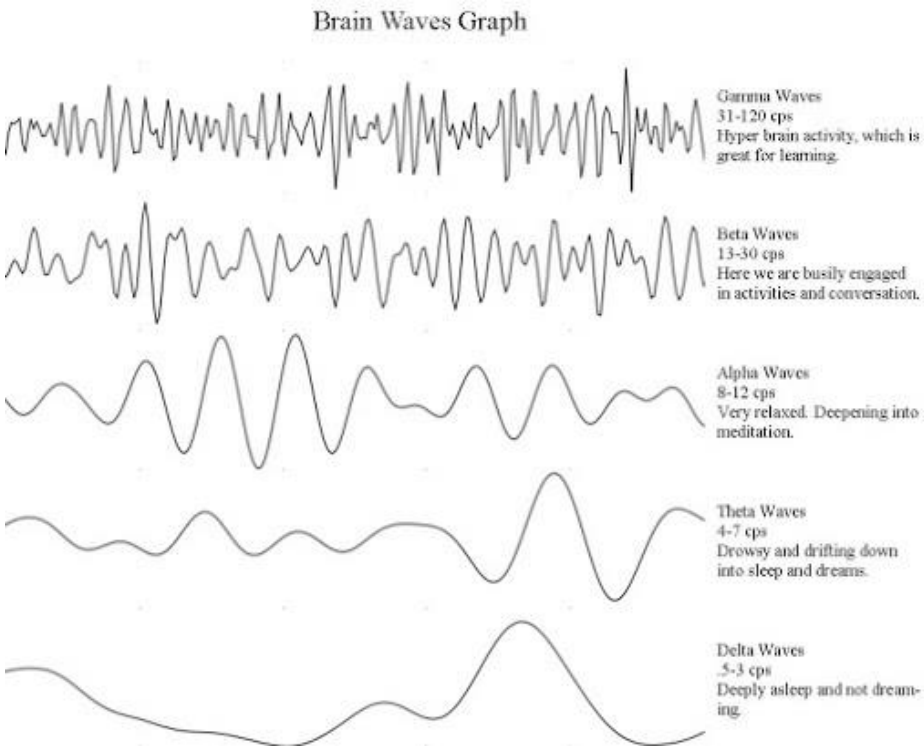
Image data


Low-Level Feature
Mid-Level Feature
High-Level Feature
Trainable Classifier

Sequential data

- Brain signal processing





Others

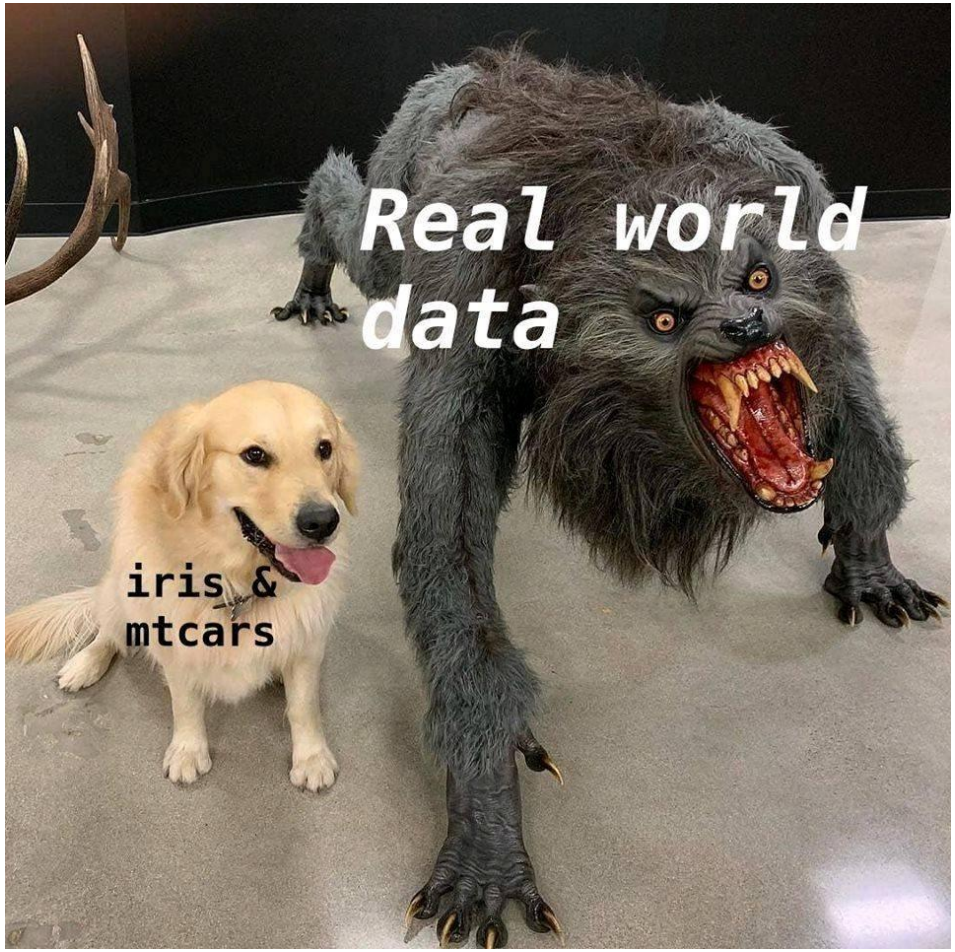
- Recommender systems
  - Netflex : 480,189 users x 17,770 movies matrix

Data Transformation

Data preprocessing

"데이터 과학의 80%는
데이터 클리닝에 소비되고,
나머지 20%는
데이터 클리닝하는 시간을
불평하는데 쓰인다."

- Kaggle 창립자·CEO Anthony Goldbloom



Data transformation

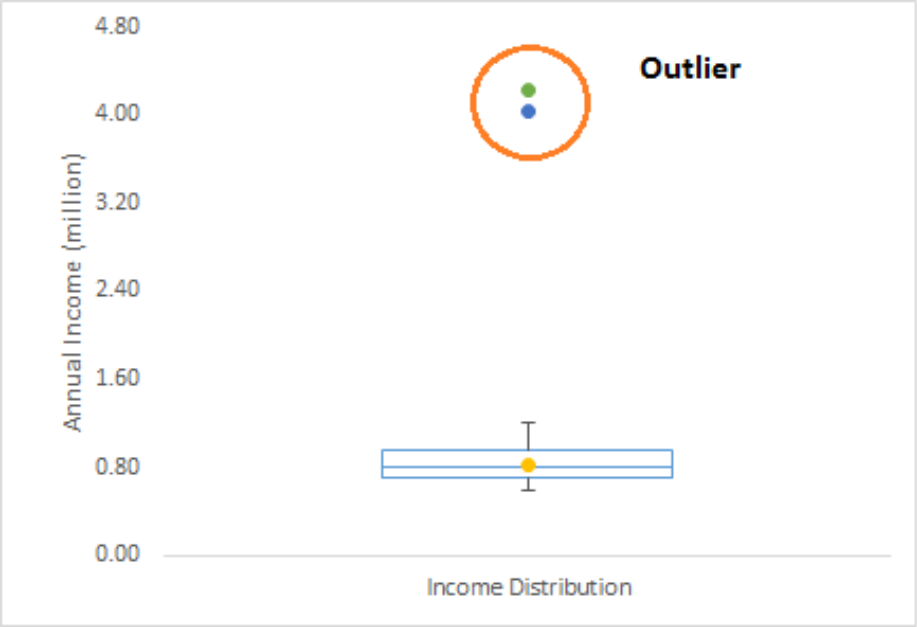
- Process data to make clear pattern for the given problem
  - Raw data → feature

- Data cleansing
  - Noise removal
  - Calibration
- Data preprocessing
  - Standardization/normalization
  - Intermediate values
- Feature representation
  - Find most suitable data representation from previous literature
- Feature selection
  - Find set of features most meaningful among data

Data cleansing

- Removes noises from data
  - Usually, a manual job
  - Find a patterns of noise by visualizing the data & removes them
  - e.g., noise from sensors at the beginning of brain wave sequences

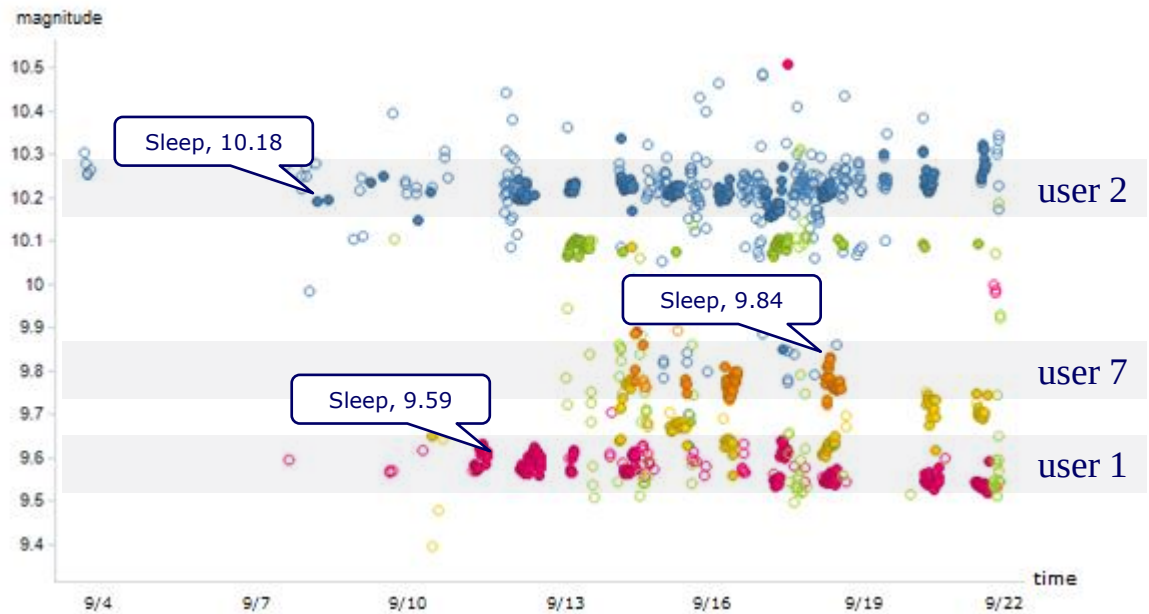
- Outlier removal
  - IQR(inter quantile range)
  - Clustering based methods



Gravity differs for devices

- Ideal : $g = 9.8m/s^2$

Accelerator for "Sleep"
about 9.5~10.3



Different range

기기	센서 MFG	측정범위(m/s^2)
삼성 갤럭시S6 (SM-G920K)	Invensense	(-39, +39)
삼성 갤럭시노트2 (SHV-E250K)	STMicroelectronics	(-19, +19)
LG G3 Beat (LG-F470K)	BOSCH	(-19, +19)

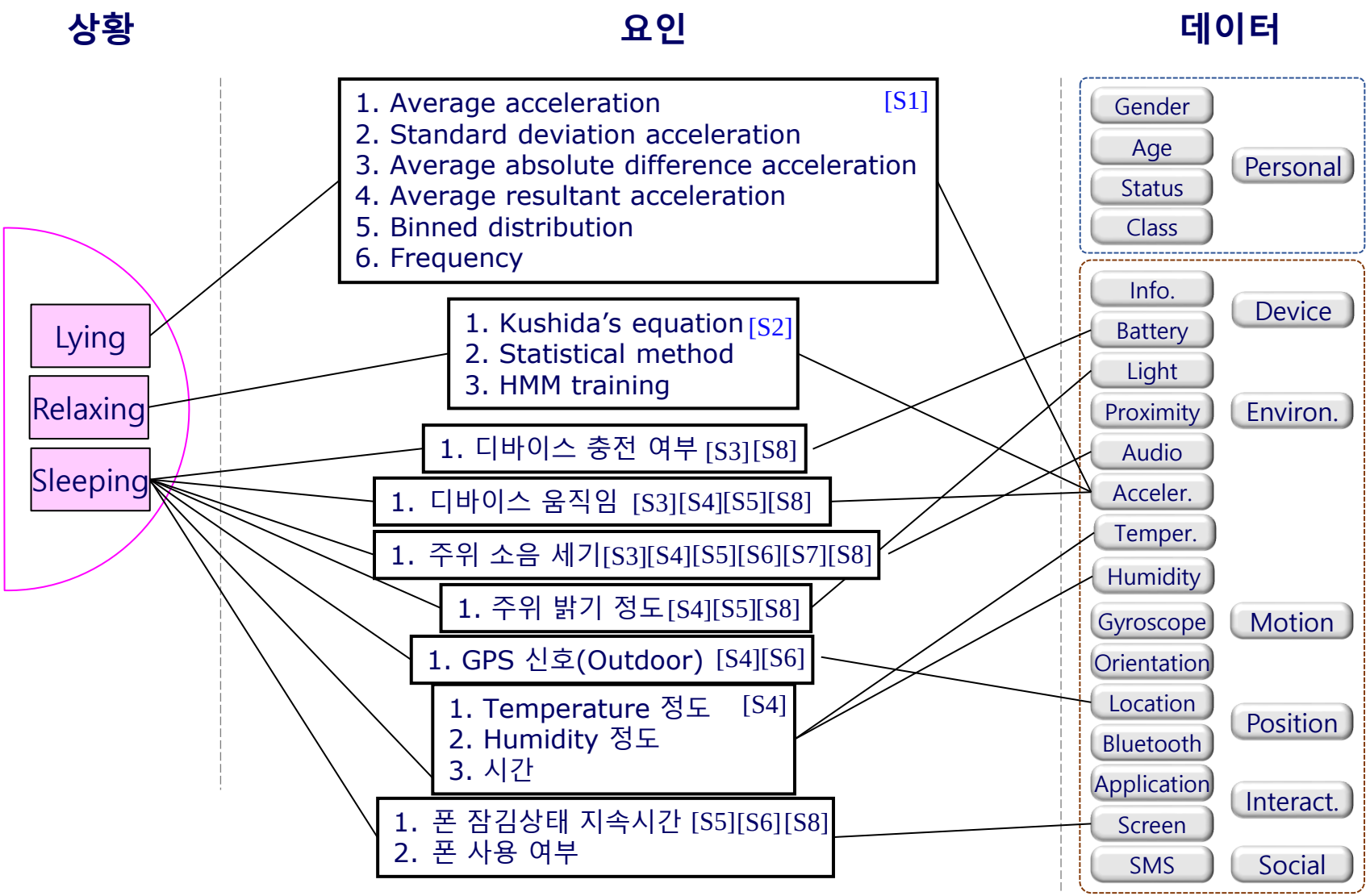
Data Preprocessing

- Data imputation
  - Fill null values in data
  - 평균값 채우기, seasonality 반영
  - ML methods : k-NN, SVD, MF, …

- Data normalization / standardization
  - Scale of attributes

- Data augmentation
  - Make additional data
  - For data imbalance or small data
  - DL methods : GAN(generative adversarial network)

- Feature selection
  - Only utilize meaningful features to enhance performance & lower costs
  - Removing features with low variance, model-based selection


상황
요인
데이터
Lying
Relaxing
Sleeping
1. Average acceleration [S1]
2. Standard deviation acceleration
3. Average absolute difference acceleration
4. Average resultant acceleration
5. Binned distribution
6. Frequency
1. Kushida's equation [S2]
2. Statistical method
3. HMM training
1. 디바이스 충전 여부 [S3][S8]
1. 디바이스 움직임 [S3][S4][S5][S8]
1. 주위 소음 세기[S3][S4][S5][S6][S7][S8]
1. 주위 밝기 정도[S4][S5][S8]
1. GPS 신호(Outdoor) [S4][S6]
1. Temperature 정도 [S4]
2. Humidity 정도
3. 시간
1. 폰 잠김상태 지속시간 [S5][S6][S8]
2. 폰 사용 여부
Gender
Age
Status
Class
Personal
Info.
Battery
Device
Light
Proximity
Environ.
Audio
Acceler.
Temper.
Humidity
Gyroscope
Motion
Orientation
Location
Position
Bluetooth
Application
Interact.
Screen
SMS
Social

Genetic algorithm (GA) based approach

- Initial feature set 및 predefined feature extraction method를 제공
- 생성된 feature에 대해 평가 및 선정
- Crossover 및 mutation을 통한 next generation 생산

✓ 초기해 및 extraction 방법론 제공 필요
✓ 긴 학습 시간 소요

Deep learning (DL) based approach

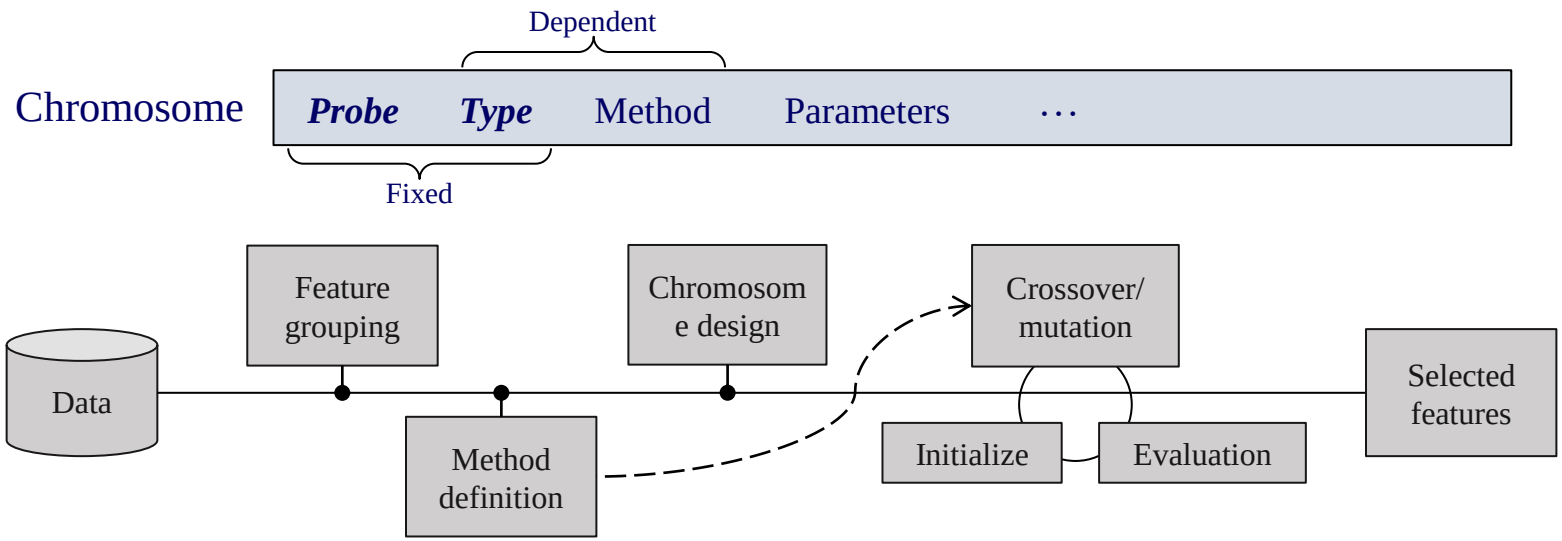
- Raw data를 기반으로 feature를 자동 생성
- Supervised 방식으로 예측 성능을 maximize하는 function 제공

✓ 다양한 형태의 데이터에 적용 사례 부재
✓ Computational cost가 높음

Unsupervised feature learning

- PCA (Principal component analysis)
- Probabilistic models, directed and undirected graphical models

✓ 사전 지식 필요
✓ 낮은 정확도



Representation learning

Shallow learning

Traditional Machine Learning Flow

Deep learning

Deep Learning Flow

ML data

- Matrix/table
- Observations(**x**, independent variables)
- Target, class(y, dependent variable)

Column/attribute/feature

Target/class

Row/
instance

ML data: example

- Student info.
- Grade

Index	ID	Name	Age
1	1111	A	22
2	1112	B	24
3	1113	C	25
4	1114	D	23

ID	Math	Programming	Science
1111	A	B	B
1112	B	B	C
1113	C	A	A
1114	B	C	C

- 과목 별 성적에 따른 학생의 차이점이 있을까?
- 학생의 성적 예측

Unstructured data

- Text data
  - Composed of strings
  - Sometimes long document
  - Word vector, w2v

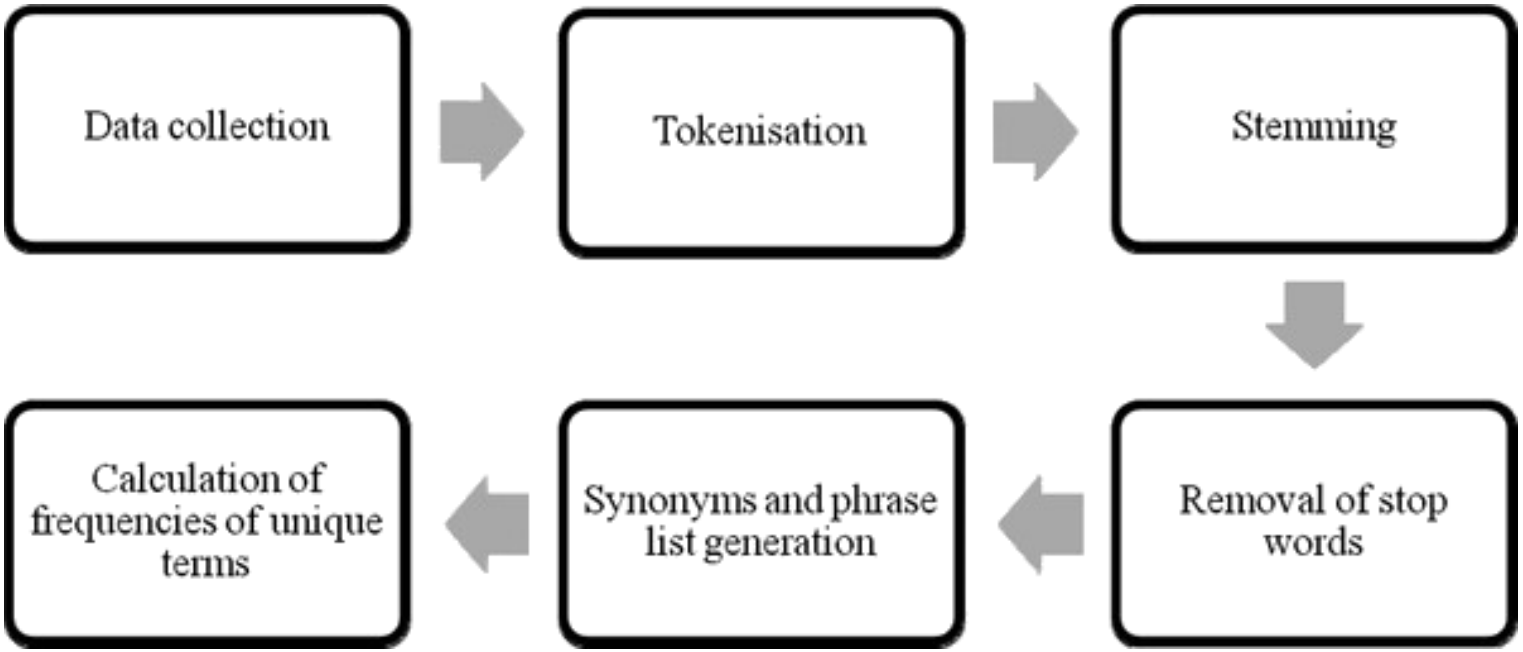
- Image / video data
  - List of vectors of (R, G, B) indicating each pixel

- Sequential data
  - Data collected according to time t
  - Signal data, audio signal, brain signal
  - Aggregated features e.g. mean/std. over 5min.
  - Few models deals with sequences e.g. HMM, RNN

- Network data
  - Node-edge

텍스트 데이터 분석

- 단어 수준으로 문서 쪼개기 – tokenization
- 단어를 일반형으로 변환 - stemming
- 불필요한 단어 제거 – stopwords
- 동의어 처리



Input

[1-3-1-3] 개인·신용·통신 정보를 안전하게 보호 개인정보 보호 체계 효율화, 개인정보보호위원회의 위상 강화 무더기 정보 이용 동의(일괄 동의)를 통한 무분별한 신용정보 활용 금지 활용 목적별, 활용 기관별로 신용정보 제공 동의를 각각 받도록 규정 목적 외, 그룹 내 무단 정보 사용에 대한 제재 강화 금융기관 정보보호 시스템 상시 평가제 도입

Tokenized

개인/신용/통신/정보를/안전하게/보호/개인정보/보호/체계/효율화/개인정보보호위원회의/위상/강화/무더기/정보/이용/동의/일괄/동의를/통한/무분별한/신용정보/활용/금지/활용/목적별/활용/기관별로/신용정보/제공/동의를/각각/받도록/규정/목적/외/그룹/내/무단/정보/사용에/대한/제재/강화/금융기관/정보보호/시스템/상시/평가제/도입

stemmed

개인,신용,통신,정보,안전,보호,개인정보,보호,체계,효율,화,개인정보,보호,위원,회의,위상,강화,무더기,정보,이용,동의,일괄,동의,를,통한,무분별,신용,정보,활용,금지,활용,목적,활용,기관,신용,정보,제공,동의,각각,받다,규정,목적,외,그룹,내,무단,정보,사용,대한,제재,강화,금융,기관,정보,보호,시스템,상시,평가,제,도입,

Document/term representation

- 텍스트 데이터의 일반적인 분석 단위는 문서(document)/단어(term)
- 문서와 단어를 벡터로 표현하여 고도화된 방법론 적용
- 가장 일반적으로 사용되는 방식은 term-document matrix를 이용한 표현법
  - 하나의 문서를 특정 단어의 등장여부/빈도수로 구성된 벡터로 표현
  - 하나의 단어를 발생한 문서로 구성된 벡터로 표현

Document space → ← Term vector space

	t_1	t_2	t_3	…	t_n
D_1	a_{11}	a_{12}	a_{13}	…	a_{1n}
D_2	a_{21}	a_{22}	a_{23}	…	a_{2n}
D_3	a_{31}	a_{32}	a_{33}	…	a_{3n}
…					
D_m	a_{m1}	a_{m2}	a_{m3}	…	a_{mn}

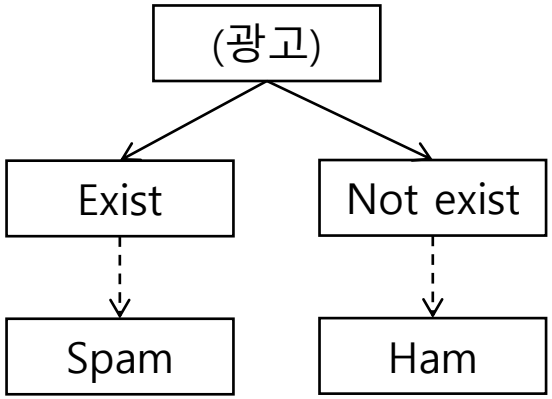
머신러닝 모델 – 예시

Email id	Existence of "(광고)"*	Spam or not**
1	1	1
2	1	1
3	1	1
4	0	0
5	0	0

- Email → 문서
- 광고 → 단어

➔ 1차원 벡터

- Spam filter
  - Determine by checking the term
  - Idea : shape of a tree
    → DT (decision tree)
  - 더 많은 단어 고려



* 1 : exist, 0 : not exist
** 1 : spam, 0 : not spam

Document/term representation

- 가장 일반적으로 사용되는 방식은 word vector 표현
  - 하나의 문서를 특정 단어의 등장여부/빈도수로 구성된 벡터로 표현
- (질문1) 아래 두 문서를 ' 엄마', '아빠', '어머니', '아버지'로 구성된 4차원 벡터로 표현하시오.

문서 1
엄마, 엄마

문서 2
어머니, 어머니

벡터 1

2	**0**	**0**	**0**

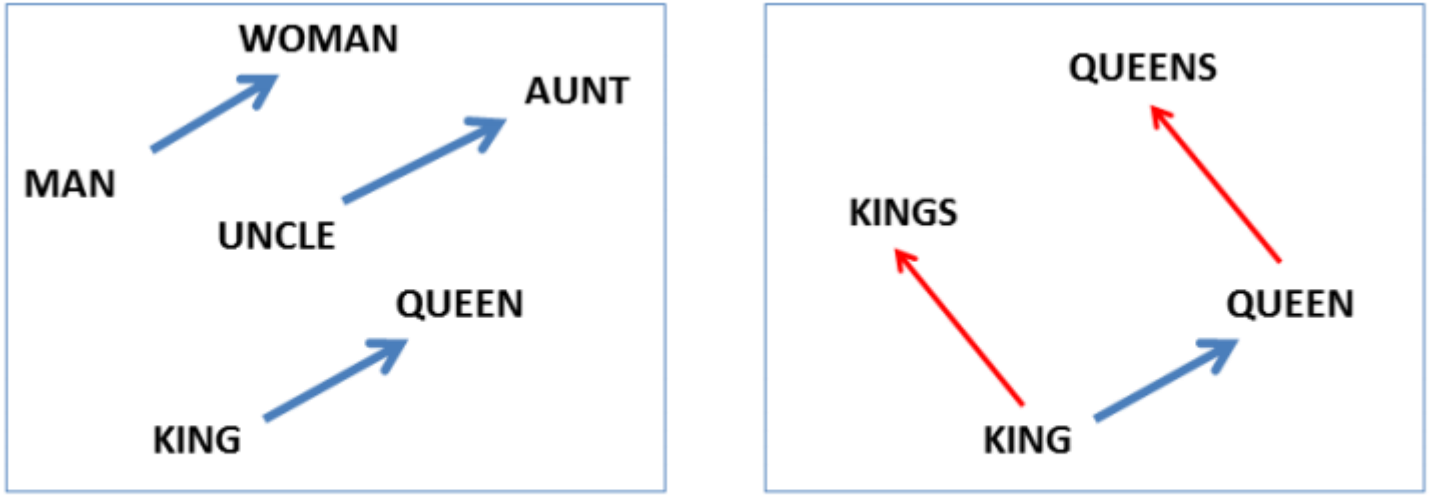
벡터 2

0	**0**	**2**	**0**

- (질문2) 일반적으로 사용될 수 있는 word vector의 차원수는?
- (질문2) 문서 1, 2는 다른 문서인가? 벡터 1, 2는 다른 벡터인가?

Document/term representation

- 의미를 담을 수 있는/차원이 너무 크지 않은 embedding 방식 필요
- 딥러닝 기반의 word2vec
- 토픽 모델링 기반의 LDA(latent Dirichlet allocation)



(Mikolov et al., NAACL HLT, 2013)

Data Exploration

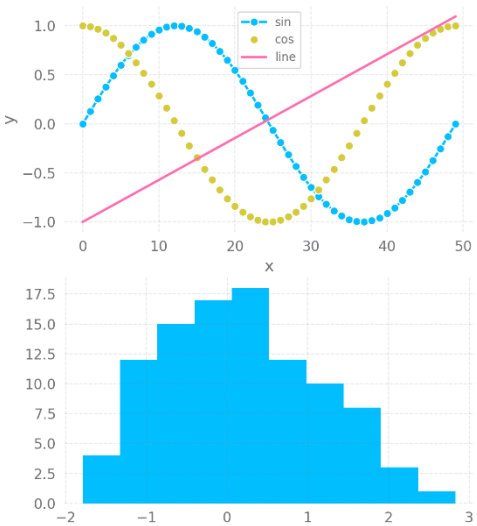
Data Exploration

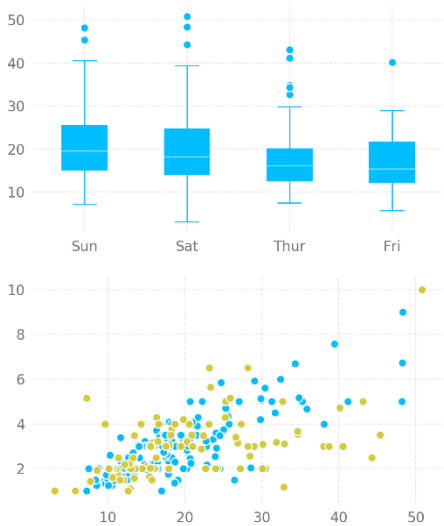
- Explores data to get insight (0) examine data one-by-one (1) data summary, (2) data visualization
- (1) data summary
  - Shape, length, statistics

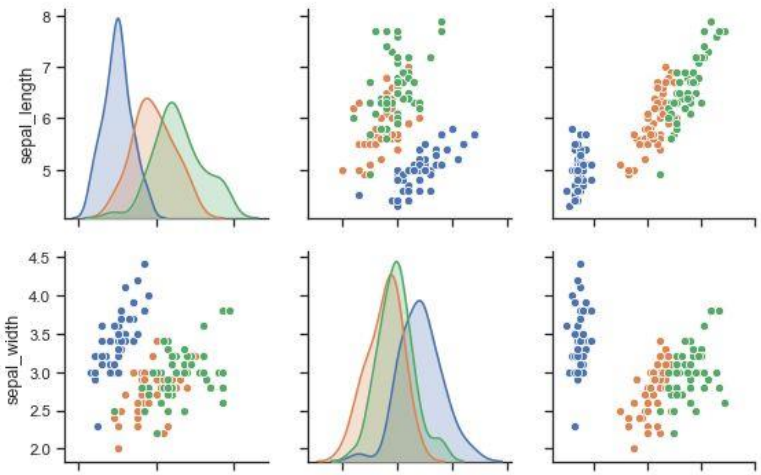
	Age	Score
count	12.000000	12.000000
mean	32.500000	73.000000
std	9.209679	17.653225
min	24.000000	44.000000
25%	25.750000	64.000000
50%	29.000000	74.000000
75%	35.250000	87.500000
max	51.000000	99.000000

Data Exploration

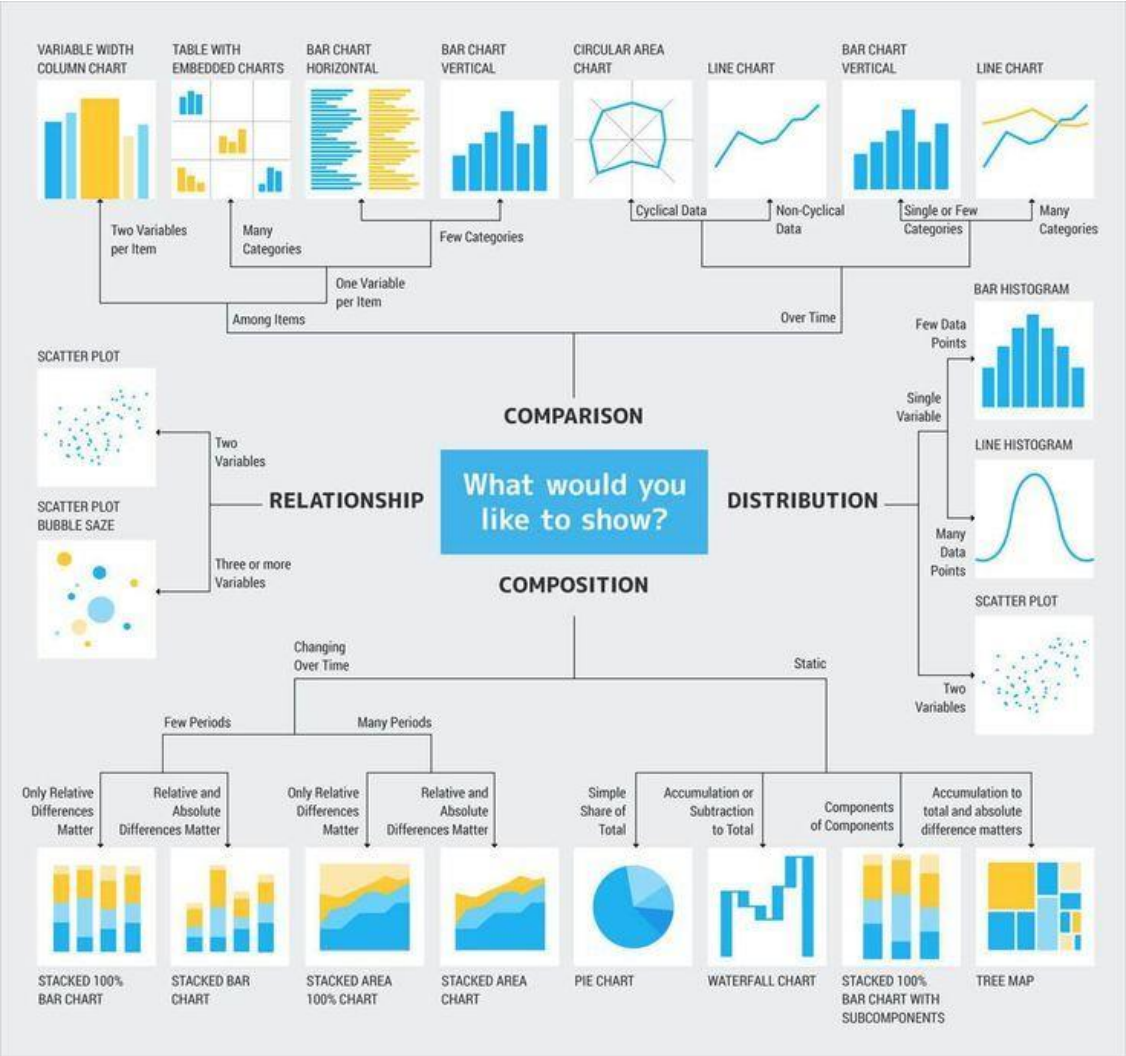
- Explores data to get insight (0) examine data one-by-one (1) data summary, (2) data visualization
- (2) data visualization
  - Histogram, bar plots, scatter plots, area plot to pie plot







Data Visualization



Data Modeling

Model training

- Select ML model that fits the problem
- Find parameters that minimize the objective function of the model

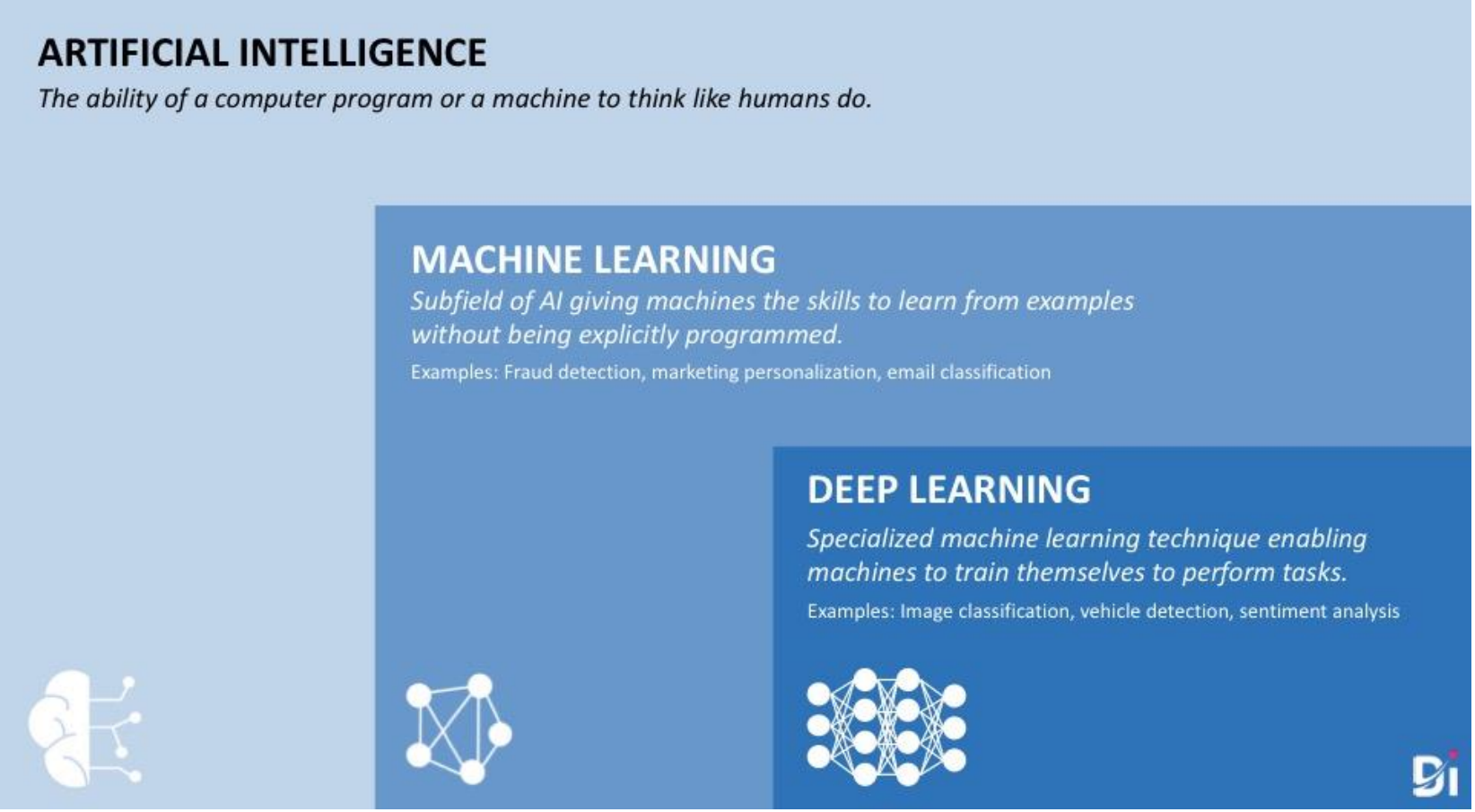
- Feature type
  - Discrete/continuous/both

- Dynamic vs. static

- Optimization
  - Loss (objective function)
  - Optimizer

Data Modeling

- ML & DL



Data Modeling

Machine Learning

Learn from data using Models & Algorithms

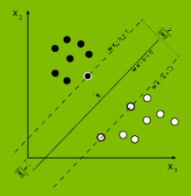
Data

Learning

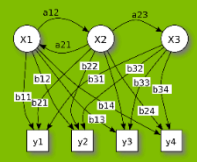
Adjusting parameters using algorithms

Model

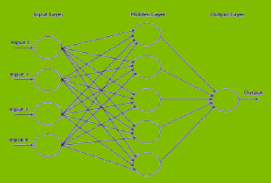
Template with parameters



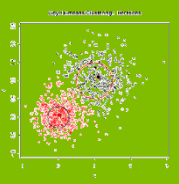
SVM



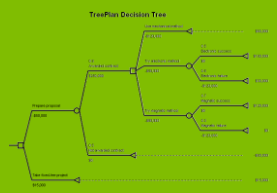
HMM



ANN



k-means



DT

Data Modeling

- Input data format – vector
  - Matrix/table
  - Observations(x, independent variables)
  - Target, class(y, dependent variable)

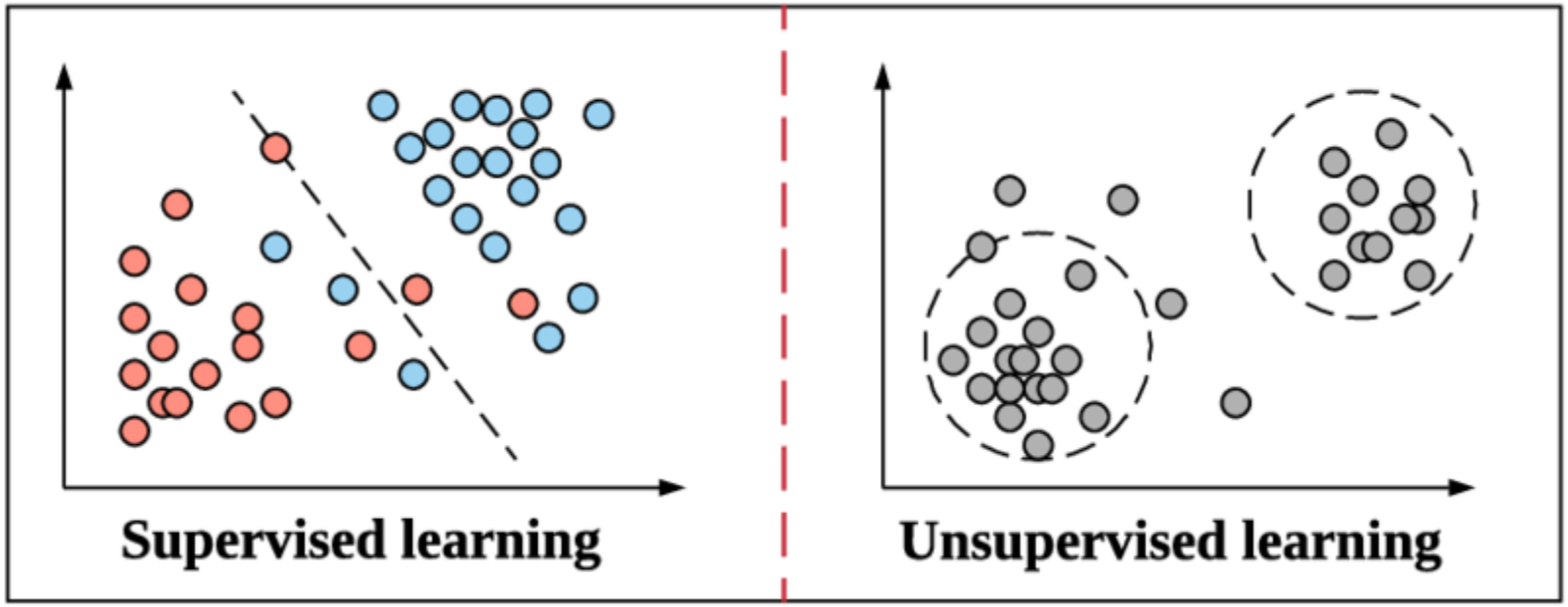
Column/attribute/feature

Target/class

Row/
instance

Models

- 데이터를 이용하여 모델을 학습
  - 데이터의 '답' 명시 여부에 따라 지도/비지도 학습으로 구분



Models

- Supervised vs. Unsupervised

	Unsupervised	Supervised
Continuous	• Clustering & Dimensionality Reduction ○ SVD ○ PCA ○ K-means	• Regression ○ Linear ○ Polynomial • Decision Trees • Random Forests
Categorical	• Association Analysis ○ Apriori ○ FP-Growth • Hidden Markov Model	• Classification ○ KNN ○ Trees ○ Logistic Regression ○ Naive-Bayes ○ SVM

Models

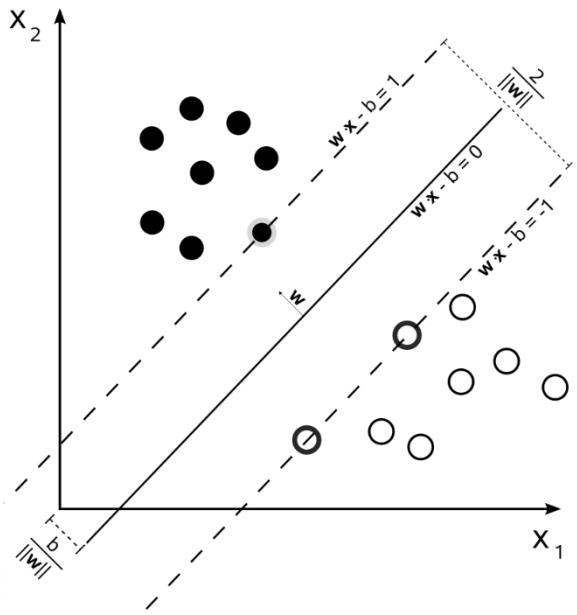
	평가요소				
	학습 소요 시간	일반적 성능	시계열 정보 처리 가능성	필요한 학습데이터양	비선형데이터 처리가능성
Support vector machine	●	●	◐	◕	●
Hidden Markov model	○	◑	●	○	●
Decision tree	◔	◑	○	◑	◔
Naïve Bayes	●	◕	◔	○	●
Neural networks	◕	◑	○	◑	●
Logistic regression	◕	○	◕	◑	●

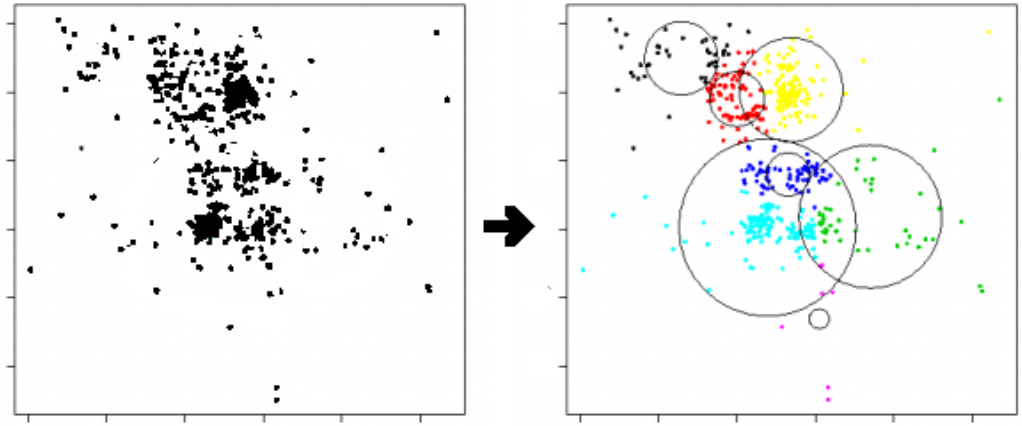
○ 약점 ⟷ ● 강점

Models

- Support Vector Machines
  - Classification method

- K-means algorithm
  - Clustering method

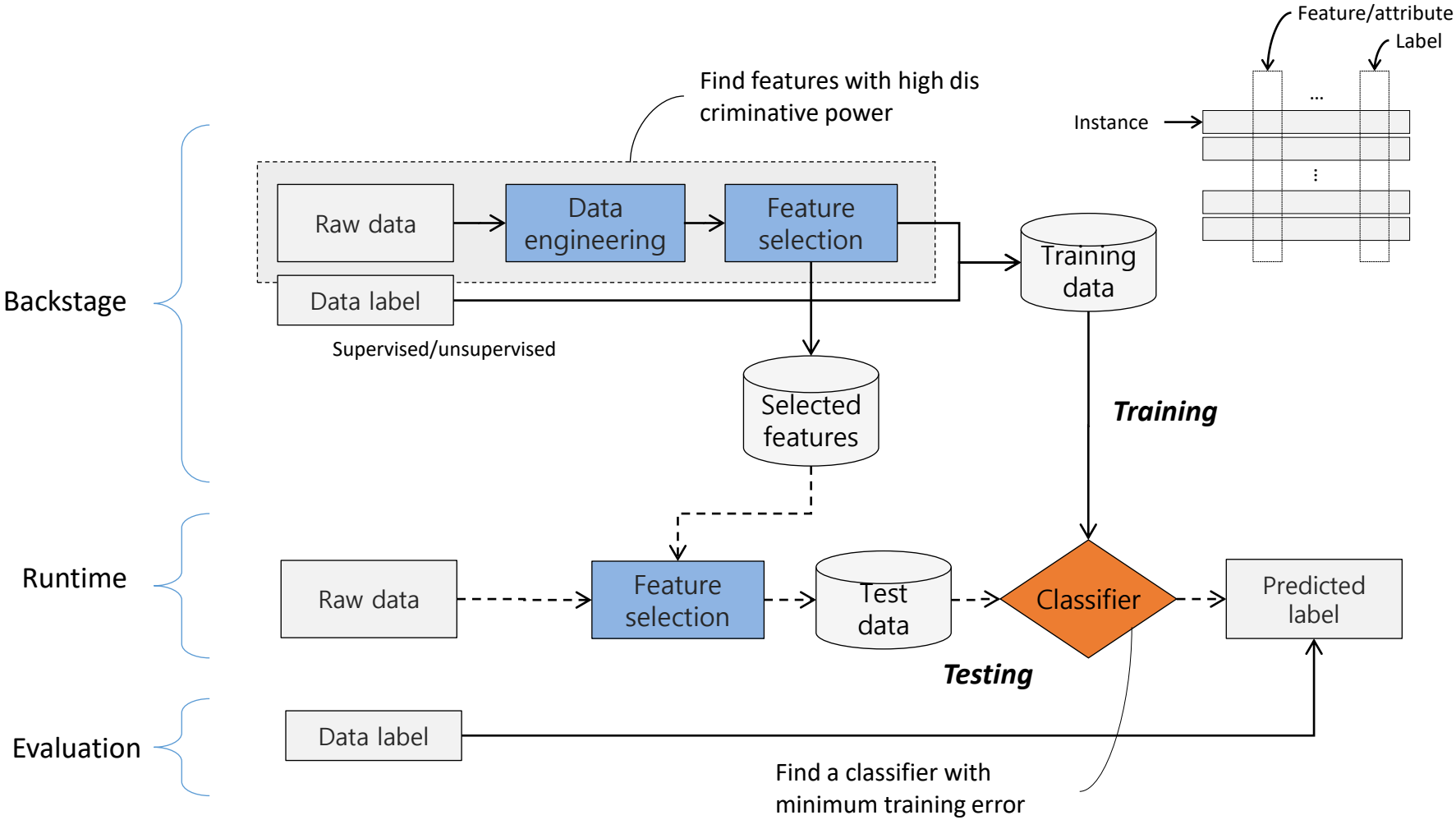




Raw Data

Clustered Data

Modeling Process

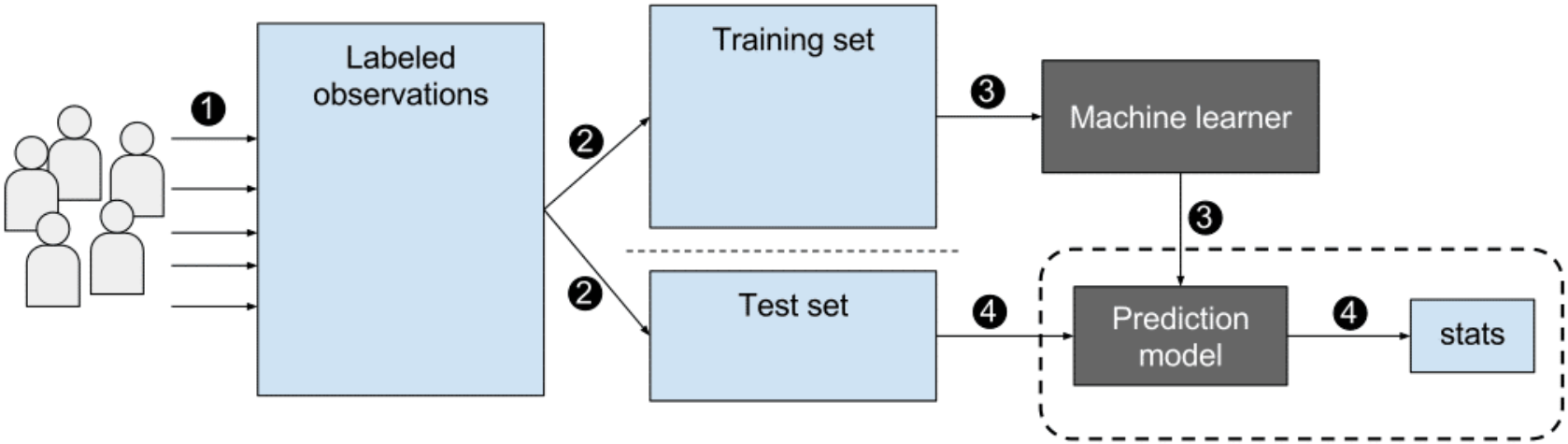


- **Which model is most suitable for the data**
- **Training the model**

Evaluation

- Data model training & evaluation process

- Evaluation metrics
  - Classification : accuracy, F1 score, recall, precision
  - Regression : RMSE, MAE, MAPE
  - Clustering : Rand index, purity, mutual information



Evaluation

- Performs evaluation of the developed model using test data compared to other state-of-the-art (SOTA) methods
  - Quantitative and qualitative evaluation

- Quantitative evaluation
  - Compares the "number" → evaluation measures
  - Test data should contain answer
  - Evaluation measures
    - Categorical : accuracy, F1, precision, recall
    - Continuous : MSE

- Qualitative evaluation
  - Check whether the model actually works – numbers could have bias
  - Visualize result samples
  - Case study

Q&A

End of Document

최예림

서울여자대학교 데이터사이언스학과

yerim.choi@swu.ac.kr